OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

9.2 Road reference line

The basic element of every road in ASAM OpenDRIVE is the road reference line. All geometry elements that describe the road shape and further properties of the road are defined along the road reference line. These properties include lanes and signals.

By definition, the road reference line runs in s-direction, while the lateral deviation of objects from the road reference line runs in t-direction. The direction of the road reference line does not indicate the driving direction.

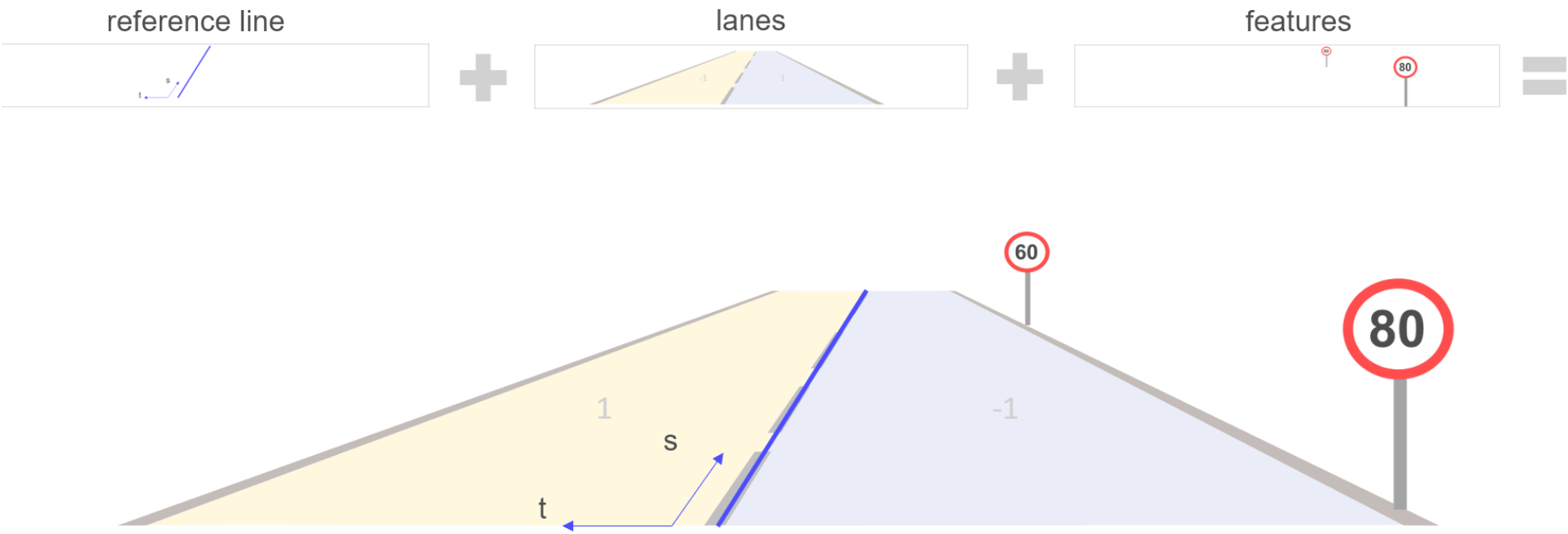


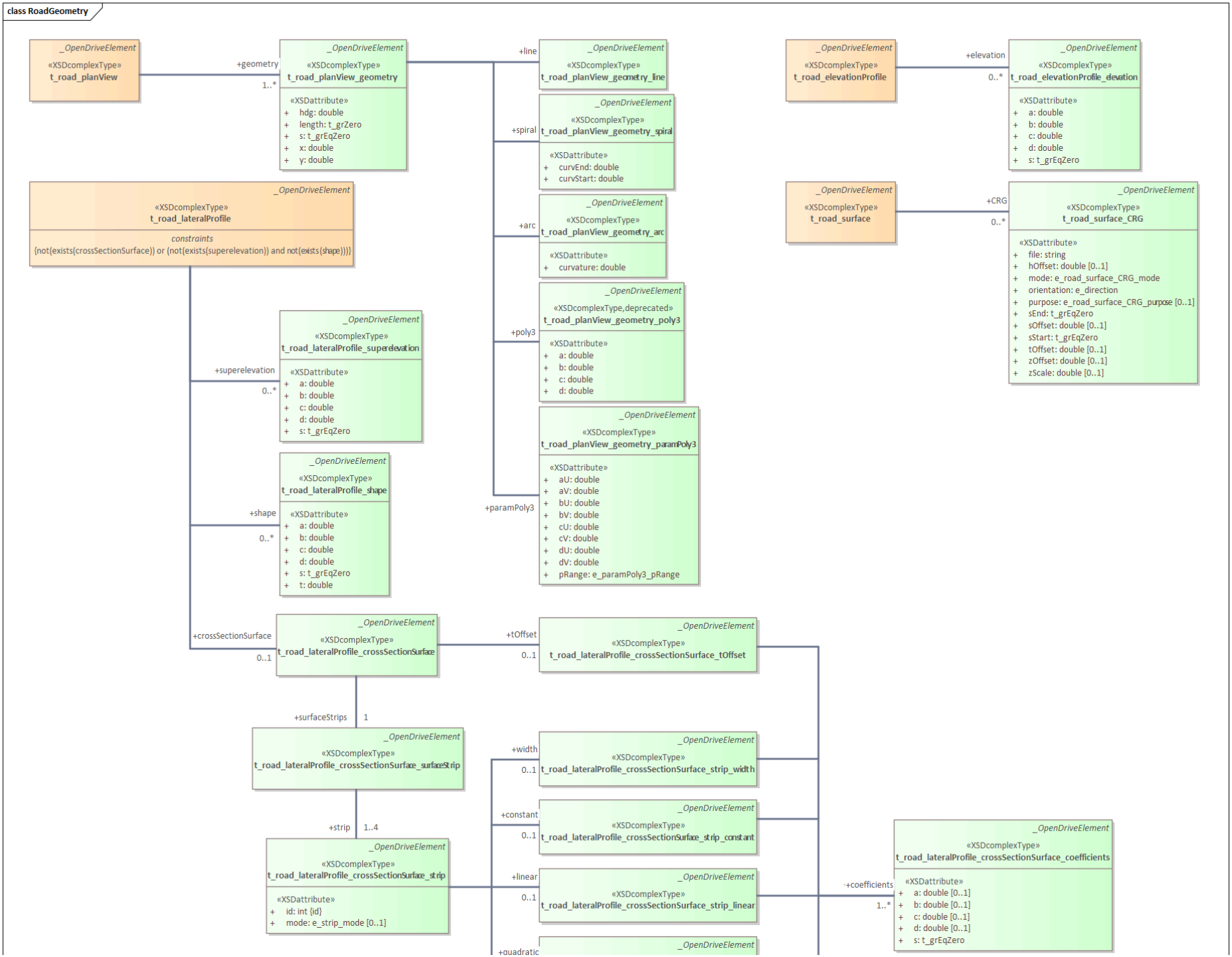
Figure 24. Individual parts of a road

Figure 24 shows the different parts of a road in ASAM OpenDRIVE.

- The road reference line

- Individual lanes of a road
- Features like signals that are placed along the road

Elements in UML model


class RoadGeometry
_OpenDriveElement
«XSDcomplexType»
t_road_planView
+geometry
1..*
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry
«XSDattribute»
+ hdg: double
+ length: t_grZero
+ s: t_grEqZero
+ x: double
+ y: double
+line
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_line
+spiral
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_spiral
«XSDattribute»
+ curvEnd: double
+ curvStart: double
+arc
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_arc
«XSDattribute»
+ curvature: double
+poly3
_OpenDriveElement
«XSDcomplexType,deprecated»
t_road_planView_geometry_poly3
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+paramPoly3
_OpenDriveElement
«XSDcomplexType»
t_road_planView_geometry_paramPoly3
«XSDattribute»
+ aU: double
+ aV: double
+ bU: double
+ bV: double
+ cU: double
+ cV: double
+ dU: double
+ dV: double
+ pRange: e_paramPoly3_pRange
_OpenDriveElement
«XSDcomplexType»
t_road_elevationProfile
+elevation
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_elevationProfile_elevation
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
_OpenDriveElement
«XSDcomplexType»
t_road_surface
+CRG
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_surface_CRG
«XSDattribute»
+ file: string
+ hOffset: double [0..1]
+ mode: e_road_surface_CRG_mode
+ orientation: e_direction
+ purpose: e_road_surface_CRG_purpose [0..1]
+ sEnd: t_grEqZero
+ sOffset: double [0..1]
+ sStart: t_grEqZero
+ tOffset: double [0..1]
+ zOffset: double [0..1]
+ zScale: double [0..1]
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile
constraints
{not(exists(crossSectionSurface)) or (not(exists(superelevation)) and not(exists(shape)))}
+superelevation
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_superelevation
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
+shape
0..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_shape
«XSDattribute»
+ a: double
+ b: double
+ c: double
+ d: double
+ s: t_grEqZero
+ t: double
+crossSectionSurface
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface
+tOffset
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_tOffset
+surfaceStrips
1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_surfaceStrip
+strip
1..4
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip
«XSDattribute»
+ id: int {id}
+ mode: e_strip_mode [0..1]
+width
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_width
+constant
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_constant
+linear
0..1
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_strip_linear
+quadratic
_OpenDriveElement
+coefficients
1..*
_OpenDriveElement
«XSDcomplexType»
t_road_lateralProfile_crossSectionSurface_coefficients
«XSDattribute»
+ a: double [0..1]
+ b: double [0..1]
+ c: double [0..1]
+ d: double [0..1]
+ s: t_grEqZero

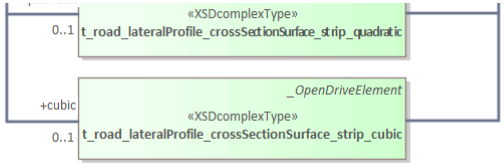


Figure 25. UML class diagram of the RoadGeometry class, including the road reference line elements

Figure 25 shows the UML class diagram of the ASAM OpenDRIVE RoadGeometry class.

`<planView>` element

In ASAM OpenDRIVE, the plan view is represented by the `<planView>` element within the `<road>` element. The `<planView>` element is a mandatory element in every `<road>` element.

```
UML class: t_road_planView
XML tag:   <planView> (Multiplicity: 1)
```

Contains geometry elements that define the layout of the road reference line in the x/y-plane (plan view).

`<geometry>` element

In ASAM OpenDRIVE, the geometry of a road reference line is represented by the `<geometry>` element within the `<planView>` element.

```
UML class: t_road_planView_geometry
XML tag:   <geometry> (Multiplicity: 1..*)
```

Table 18. Attributes of the <geometry> element

Name	Type	Use	Unit	Description
`hdg`	double	required	rad	Start orientation (inertial heading)
`length`	t_grZero	required	m	Length of the element's reference line

Name	Type	Use	Unit	Description
s	t_grEqZero	required	m	s-coordinate of start position
x	double	required	m	Start position (x inertial)
y	double	required	m	Start position (y inertial)

Rules

The following rules apply to road reference lines:

- asam.net:xodr:1.4.0:road.geometry.refline_exists: Each road shall have a road reference line.
- asam.net:xodr:1.4.0:road.geometry.only_one_refline: There shall be only one road reference line per road.
- The road reference line usually runs in the center of the road but may be laterally offset.
- asam.net:xodr:1.4.0:road.geometry.elem_asc_order: `<geometry>` elements shall be defined in ascending order along the road reference line according to the s-coordinate.
- asam.net:xodr:1.4.0:road.geometry.one_geom_elem_per_spec: One `<geometry>` element shall contain only one element that further specifies the geometry of the road.
- asam.net:xodr:1.7.0:road.geometry.contact_point: If two roads are connected without a junction, the road reference line of a new road shall always begin at the `<contactPoint>` element of its successor or predecessor road. The road reference lines may be directed in opposite directions.
- asam.net:xodr:1.4.0:road.geometry.refline_no_gaps: A road reference line shall have no gaps.
- asam.net:xodr:1.4.0:road.geometry.refline_no_kinks: A road reference line should have no kinks.
- The s-value of each `<geometry>` shall be the sum of all `<geometry>` lengths prior

Related topics

- Section 9.3, "Straight line"
- Section 9.4, "Spiral"
- Section 9.5, "Arc"
- Section 10.1, "Introduction to roads"
- Section 11.1, "Introduction to lanes"